

CODEALPHA CYBER SECURITY INTERNSHIP

TASK 1: BASIC NETWORK SNIFFER

Submitted By

Name: Deney Dasari

Internship Domain

Cyber Security

Task Name

Basic Network Sniffer

1. Objective

The objective of this project is to develop a Basic Network Sniffer using Python and the Scapy library. The tool captures network packets flowing through the system and displays important packet information such as source IP address, destination IP address, protocol type, and payload data. This helps in understanding network communication and packet structures.

2. Tools and Technologies Used

- Python 3.14
 - Scapy Library
 - Npcap
 - Visual Studio Code
 - Windows 11
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3. Project Description

A Network Sniffer is a tool used to monitor and analyze network traffic. In this project, Scapy was used to capture TCP packets and extract useful information from each packet.

The program captures packets and displays:

- Source IP Address
- Destination IP Address
- Protocol Number
- Payload Data

The packet information helps in understanding how devices communicate across a network.

4. Source Code

```
from scapy.all import *

def packet_callback(packet):
    print("\n=== Packet Captured ===")

    if packet.haslayer(IP):
        print("Source IP:", packet[IP].src)
        print("Destination IP:", packet[IP].dst)
        print("Protocol:", packet[IP].proto)

    if packet.haslayer(Raw):
        print("Payload:")
        print(packet[Raw].load)

sniff(filter="tcp", prn=packet_callback, store=False, count=20)
```

5. Procedure

1. Installed Python and Scapy library.
 2. Installed Npcap for packet capturing support.
 3. Created a Python script using Scapy.
 4. Configured the program to capture TCP packets.
 5. Executed the script in Visual Studio Code.
 6. Generated network traffic by browsing websites.
 7. Captured and analyzed network packets.
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6. Output

The program successfully captured network packets and displayed:

- Source IP Address
- Destination IP Address
- Protocol Information

- Payload Content

Sample Output:

Source IP: 192.168.0.101

Destination IP: 40.79.163.155

Protocol: 6 (TCP)

Payload: Captured network data

7. Screenshots

Insert the following screenshots:

1. Source Code Screenshot
 2. Successful Packet Capture Output Screenshot
 3. Network Traffic Monitoring Screenshot
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8. Learning Outcomes

Through this project, the following concepts were learned:

- Basics of Packet Sniffing
 - TCP/IP Communication
 - Network Traffic Analysis
 - Scapy Packet Manipulation
 - Network Security Monitoring Fundamentals
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9. Conclusion

The Basic Network Sniffer was successfully developed using Python and Scapy. The application captured live network packets and displayed relevant packet details including source IP address, destination IP address, protocol information, and payload data. This project provided practical experience in packet analysis and improved understanding of network communication and cybersecurity fundamentals.